

Dual-Mode Soil Moisture–Based Irrigation System Using Digital Logic

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Abstract—This project presents a dual-mode irrigation system that automates plant watering using digital logic and an Arduino Uno. The system operates in either manual or automatic mode by combining user input with soil moisture data through a Boolean logic expression. The analog moisture reading is converted into a binary condition (M) by comparing the ADC value with a predefined threshold. The system was validated in Tinkercad before implementation on physical hardware. Testing demonstrated reliable operation in both manual and automatic modes, showing the practical application of digital logic and embedded programming in an automated irrigation system.

Keywords—Arduino Uno, digital logic, Boolean algebra, embedded systems, irrigation system, relay module, soil moisture sensor.

I. INTRODUCTION

Automated irrigation systems improve water efficiency by supplying water only when soil conditions require it. Unlike timer-based irrigation, sensor-based systems respond to actual environmental conditions, reducing unnecessary watering while simplifying plant maintenance.

This project develops a dual-mode irrigation system using an Arduino Uno, soil moisture sensor, and relay-controlled water pump. Boolean logic determines whether the pump should operate in manual or automatic mode, demonstrating the practical application of digital logic in an embedded automation system.

II. SYSTEM DESIGN

A. System Overview

The irrigation system uses three primary inputs: a mode selection switch (A), a manual push button (B), and the soil moisture condition (M). The Arduino evaluates these inputs using the Boolean expression

$$P = (A \cdot B) + (\bar{A} \cdot M)$$

where P represents the pump output.

When manual mode is selected, the pump is activated by the push button. In automatic mode, the Arduino monitors the soil moisture sensor and activates the pump whenever dry soil is detected. The Boolean output controls a relay module that switches power to the DC water pump.

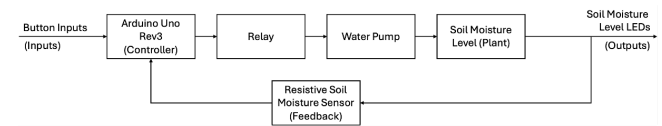


Fig. 1. Functional overview of the dual-mode irrigation system.

B. Hardware Components

The hardware consists of an Arduino Uno, a resistive soil moisture sensor, a relay module, a DC water pump, a push button, a mode selection switch, an external power supply, and a breadboard with jumper wires. The Arduino processes sensor readings and user inputs before controlling the relay that supplies power to the pump.

TABLE I
Hardware Components

Part	Description
Arduino Uno Rev3	Microcontroller board that processes sensor input and controls system outputs
Breadboard and EE 2501 Parts Kits	Used for building and connecting the circuit.
Power Supply	External power source used to operate the pump.
Relay Module	Electromechanical switching module used to control power delivery to the water pump.

Soil Moisture Sensor	Resistive analog sensor that measures soil water content through conductivity changes.
Water Pump	Actuator that delivers water to the soil when activated by the system.

C. Sensor Signal Processing

The soil moisture sensor produces an analog voltage that varies with the conductivity of the surrounding soil. The Arduino converts this signal into a digital value using its built-in Analog-to-Digital Converter (ADC), which represents sensor readings on a scale from 0 to 1023. The measured value is then compared with a predefined moisture threshold to determine whether the soil is dry or sufficiently moist. This comparison converts the analog sensor measurement into the binary moisture condition (M) used by the Boolean logic during automatic operation.

D. Digital Logic Design

The Boolean expression $P = (A \cdot B) + (\bar{A} \cdot M)$ combines the operating mode, manual input, and soil moisture condition to determine relay activation. This allows a single logic structure to support both manual and automatic watering modes.

III. IMPLEMENTATION

A. Development Process

Development began by testing the soil moisture sensor to obtain stable readings under both wet and dry soil conditions. Initial testing showed minor fluctuations, so the moisture threshold was adjusted to provide consistent separation between wet and dry soil. After calibration, manual operation was implemented by connecting a push button to the Arduino and relay module, verifying communication between the microcontroller and external hardware.

Before assembling the physical prototype, the complete circuit design was modeled and tested in Tinkercad. The electrical schematic was first created to verify circuit connections and logic implementation, followed by a full circuit simulation to validate system operation.

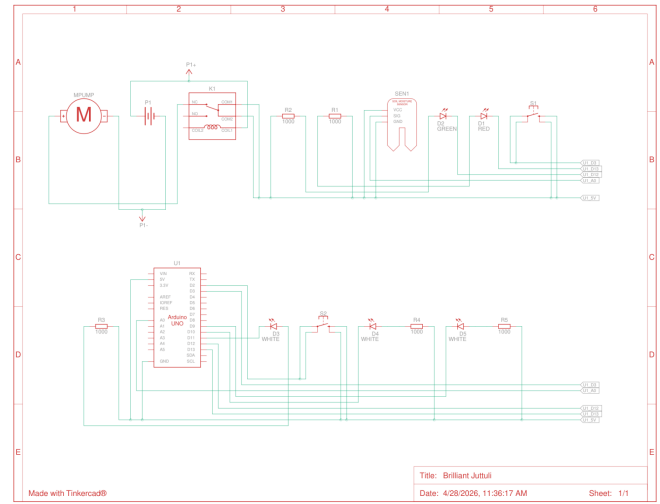


Fig. 2. Electrical schematic of the dual-mode irrigation system created in Tinkercad.

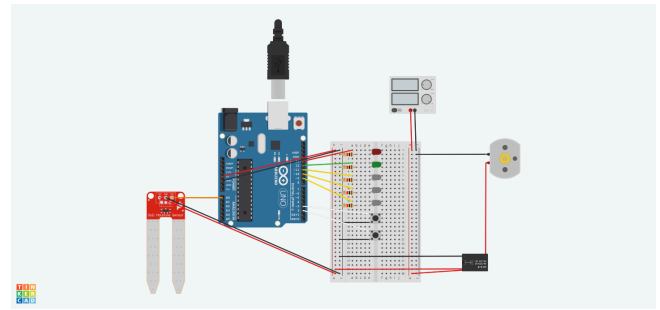


Fig. 3. Tinkercad simulation of the irrigation system.

Note: Tinkercad does not include the same relay module or water pump used in the physical prototype. Therefore, a DC motor and an SPDT relay were used as functional equivalents during simulation.

An early version of the project used timer-based irrigation. However, testing showed that the pump operated regardless of soil conditions, resulting in unnecessary watering. The timer was replaced with a moisture-based approach that activates the pump only when the soil moisture reading falls below the predefined threshold. The completed hardware implementation is shown in Fig. 4.

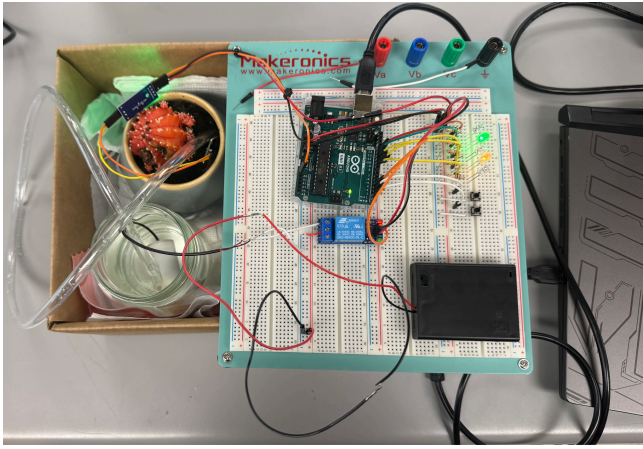


Fig. 4. Final hardware implementation of the dual-mode irrigation system.

B. Logic Flow

The Arduino continuously reads the mode selection switch, manual push button, and soil moisture sensor. The resulting input conditions are evaluated using the Boolean expression to determine relay activation. The logic gate implementation of the Boolean expression is shown in Fig. 5.

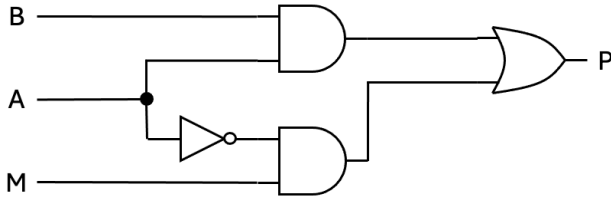


Fig. 5. Logic gate diagram of the dual-mode irrigation system.

C. Truth Table

The Boolean logic was verified using the truth table shown in Table II.

TABLE II
Truth Table for Pump Operation

A	B	M	P
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0

1	0	1	0
1	1	0	1
1	1	1	1

where A represents the mode selection (1 = Manual, 0 = Automatic), B represents the manual push button input, M represents the soil moisture condition (1 = dry soil), and P represents the pump output. The truth table verifies the Boolean expression for all possible input combinations. In manual mode ($A = 1$), the pump is controlled by the push button, while in automatic mode ($A = 0$), the pump responds to the soil moisture condition.

D. Software Implementation

The Boolean logic was implemented in the Arduino program using standard logical operators.

```
bool A = manualMode;
bool B = (digitalRead(MANUAL_BUTTON) == LOW);
bool M = (analogRead(SENSOR_PIN) >= threshold);
```

```
bool P = (A && B) || (!A && M);
```

```
digitalWrite(RELAY_PIN, !P);
```

This implementation directly corresponds to the Boolean expression shown in Fig. 5 and satisfies the required digital logic complexity using one NOT gate, two AND gates, and one OR gate.

IV. RESULTS AND DISCUSSION

The completed irrigation system met all functional requirements and operated reliably in both manual and automatic modes. During testing, the relay consistently activated the water pump according to the implemented Boolean logic. In automatic mode, the system responded to changes in soil moisture by activating the pump only when dry soil was detected, while manual mode allowed the user to control the pump directly using the push button.

Replacing the original timer-based approach with moisture-based operation improved system performance by allowing watering decisions to respond to actual soil conditions instead of fixed time intervals. During hardware assembly, creating

a stable electrical connection between the relay module and water pump proved challenging. Soldering the pump wires to jumper wires provided a secure and reliable connection for the final prototype.

Overall, the prototype successfully demonstrated the use of Boolean logic for reliable manual and automatic irrigation control.

V. CONCLUSION

This project demonstrated how Boolean logic and embedded programming can be applied to create a dual-mode irrigation system. The Arduino-based design combined manual input and soil moisture sensing to control a relay-operated water pump.

Testing confirmed reliable operation in both manual and automatic modes, with accurate responses to user input and soil conditions.

Future improvements could include adding a display to show real-time soil moisture readings and the current operating mode, integrating wireless monitoring through Wi-Fi or Bluetooth, and expanding the system to support multiple moisture sensors for larger irrigation applications. These improvements would enhance usability while maintaining the digital logic foundation of the system.

REFERENCES

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